Q1	Install / configure the VM and apply the latest security patches	5 Marks

Objective: Bring a Linux virtual machine up to date so that known vulnerabilities in installed packages are closed before any other hardening.

Step-by-Step Algorithm

Step	Phase	Action	Command / Tool	What You Should See
1	SETUP	Take a clean snapshot of the VM so you can roll back if needed.	`VirtualBox / VMware`	Snapshot saved
2	CONFIGURE	Refresh the package index to learn what updates are available.	`sudo apt update`	Package lists updated
3	CONFIGURE	Upgrade all installed packages to apply security patches.	`sudo apt upgrade -y`	Packages upgraded
4	TEST	Confirm nothing is left upgradable.	`apt list --upgradable`	List is empty
5	REPORT	Record the kernel/OS version as evidence of patch level.	uname -r ; lsb_release -a	Version noted

Phase key: ● SETUP · ● CONFIGURE · ● EXECUTE · ● TEST · ● ANALYSE · ● REPORT · ● FINISH

Worked Example (with sample values & output)

Applying the steps to lab-vm

Running the update and upgrade on the fresh VM shows several security updates being applied. Afterwards the upgradable list is empty, confirming the system is patched.

Command	Purpose	Expected result
sudo apt update	Refresh package index	Lists refreshed from mirrors
sudo apt upgrade -y	Install available updates	Security patches applied
apt list --upgradable	Verify none remain	(empty) after upgrade

Sample output:

```
$ apt list --upgradable
Listing... Done
$   (no packages listed — system is up to date)
```

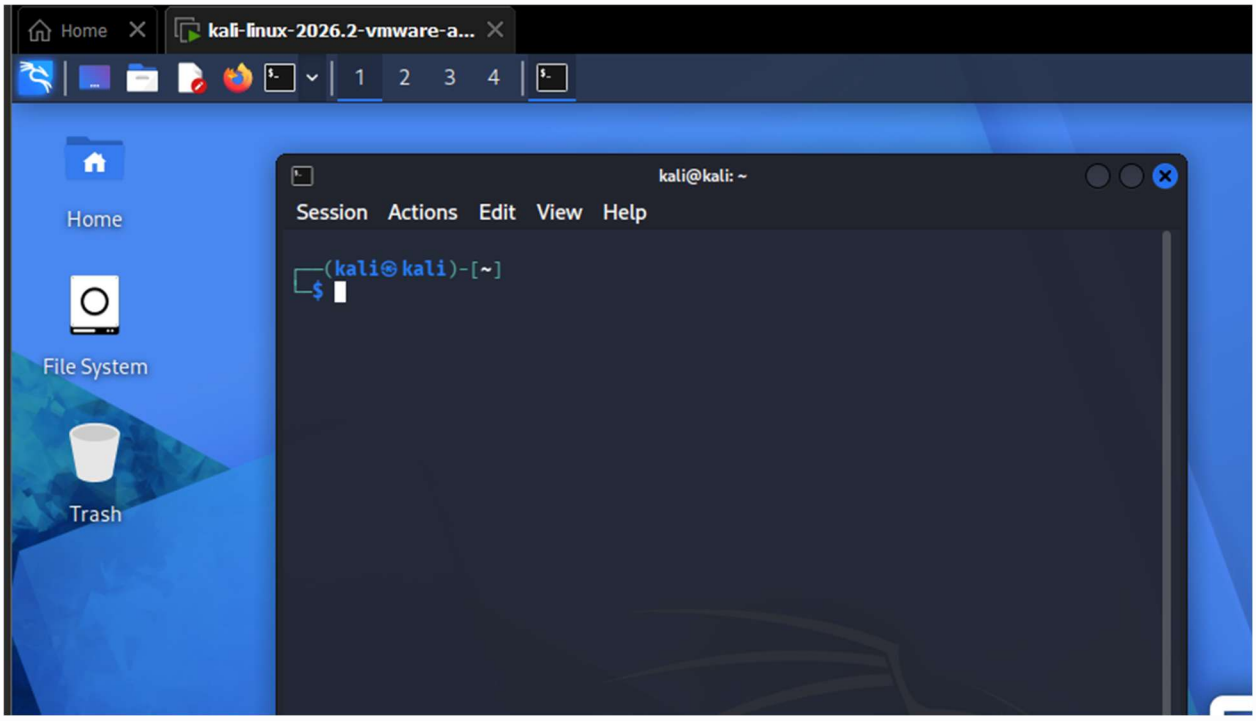
Reusable Template

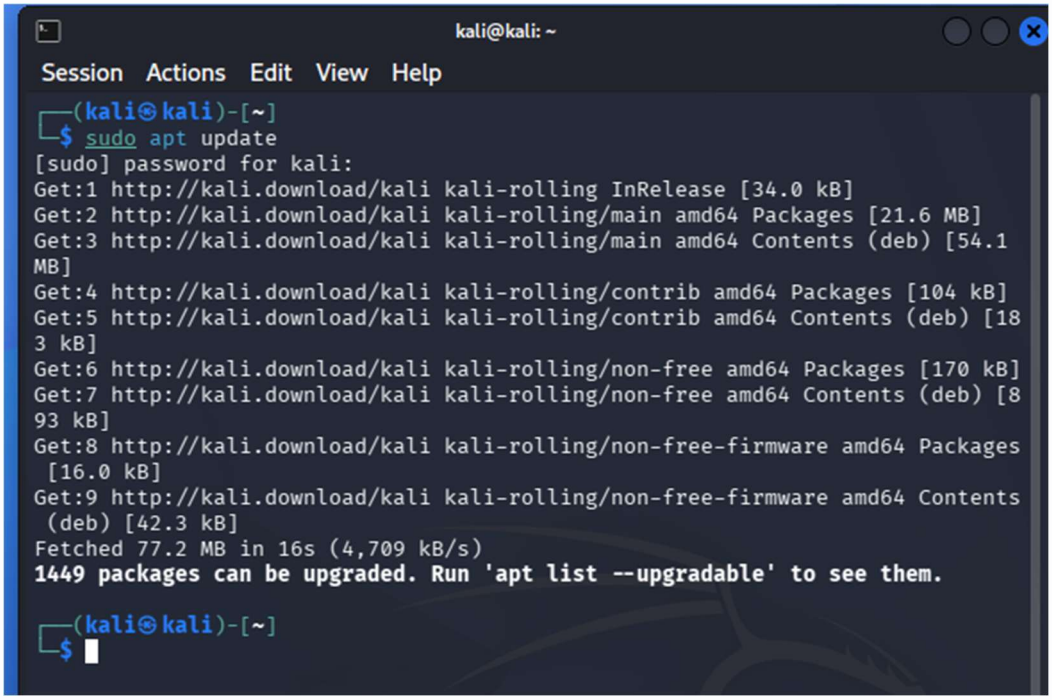
Patch-evidence template

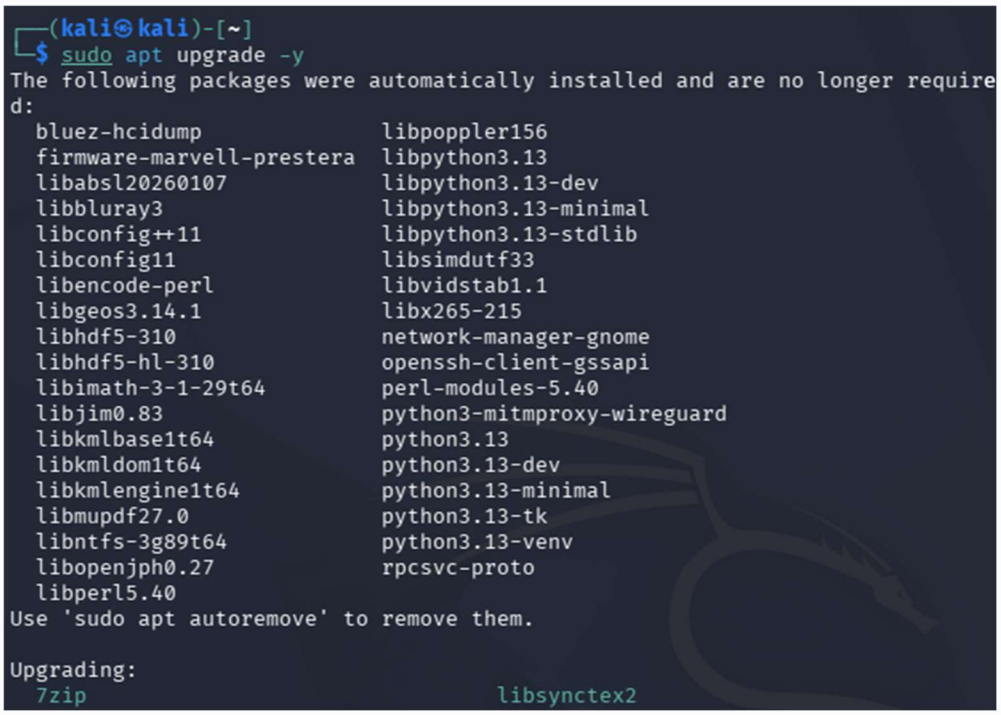
Item	Value
OS version	Kali GNU/Linux Rolling 2026.3 (from lsb_release -a)
Kernel version	7.1.5+kali-amd64 (from uname -r)
Date patched	10/06/2026
Updates applied	1449 packages

Your Output / Screenshot (paste result here):






kali@kali: ~
Session Actions Edit View Help
(kali㉿kali)-[~]
$ sudo apt update
[sudo] password for kali:
Get:1 http://kali.download/kali kali-rolling InRelease [34.0 kB]
Get:2 http://kali.download/kali kali-rolling/main amd64 Packages [21.6 MB]
Get:3 http://kali.download/kali kali-rolling/main amd64 Contents (deb) [54.1 MB]
Get:4 http://kali.download/kali kali-rolling/contrib amd64 Packages [104 kB]
Get:5 http://kali.download/kali kali-rolling/contrib amd64 Contents (deb) [183 kB]
Get:6 http://kali.download/kali kali-rolling/non-free amd64 Packages [170 kB]
Get:7 http://kali.download/kali kali-rolling/non-free amd64 Contents (deb) [893 kB]
Get:8 http://kali.download/kali kali-rolling/non-free-firmware amd64 Packages [16.0 kB]
Get:9 http://kali.download/kali kali-rolling/non-free-firmware amd64 Contents (deb) [42.3 kB]
Fetched 77.2 MB in 16s (4,709 kB/s)
1449 packages can be upgraded. Run 'apt list --upgradable' to see them.
(kali㉿kali)-[~]
$


(kali㉿kali)-[~]
$ sudo apt upgrade -y
The following packages were automatically installed and are no longer required:
bluez-hcidump
firmware-marvell-prestera
libabsl20260107
libbluray3
libconfig++11
libconfig11
libencode-perl
libgeos3.14.1
libhdf5-310
libhdf5-hl-310
libimath-3-1-29t64
libjim0.83
libkmlbase1t64
libkmldom1t64
libkmlengine1t64
libmupdf27.0
libntfs-3g89t64
libopenjph0.27
libperl5.40
libpoppler156
libpython3.13
libpython3.13-dev
libpython3.13-minimal
libpython3.13-stdlib
libsimdutf33
libvidstab1.1
libx265-215
network-manager-gnome
openssh-client-gssapi
perl-modules-5.40
python3-mitmproxy-wireguard
python3.13
python3.13-dev
python3.13-minimal
python3.13-tk
python3.13-venv
rpcsvc-proto
Use 'sudo apt autoremove' to remove them.
Upgrading:
7zip
libsynctex2

```
Unpacking system-config-printer-udev (1.5.18-6) over (1.5.18-4)…
Preparing to unpack …/1042-tpm-udev_4.1.3-7_all.deb…
Unpacking tpm-udev (4.1.3-7) over (4.1.3-6)…
Selecting previously unselected package systemd-tpm.
Preparing to unpack …/1043-systemd-tpm_261.2-1_amd64.deb…
Unpacking systemd-tpm (261.2-1)…
Preparing to unpack …/1044-tdb-tools_2%3a1.4.15+samba4.24.6+dfsg-1_amd64.deb…
Unpacking tdb-tools (2:1.4.15+samba4.24.6+dfsg-1) over (2:1.4.15+samba4.24.3+
dfsg-1)…
Preparing to unpack …/1045-tex-gyre_20180621-8_all.deb…
Unpacking tex-gyre (20180621-8) over (20180621-7)…
Preparing to unpack …/1046-texlive-pictures_2026.20260911-1_all.deb…
Unpacking texlive-pictures (2026.20260911-1) over (2026.20260527-3)…
Preparing to unpack …/1047-texlive-latex-base_2026.20260911-1_all.deb…
Unpacking texlive-latex-base (2026.20260911-1) over (2026.20260527-3)…
Preparing to unpack …/1048-texlive-latex-extra_2026.20260911-1_all.deb…
Unpacking texlive-latex-extra (2026.20260911-1) over (2026.20260527-1)…
^[[B^[[BPreparing to unpack …/1049-texlive-latex-recommended_2026.20260911-1_
all.deb…
Unpacking texlive-latex-recommended (2026.20260911-1) over (2026.20260527-3)…
Preparing to unpack …/1050-texlive-fonts-recommended_2026.20260911-1_all.deb…
Unpacking texlive-fonts-recommended (2026.20260911-1) over (2026.20260527-3)…
Preparing to unpack …/1051-texlive-base_2026.20260911-1_all.deb…
Unpacking texlive-base (2026.20260911-1) over (2026.20260527-3)…
Preparing to unpack …/1052-texlive-plain-generic_2026.20260911-1_all.deb…
Unpacking texlive-plain-generic (2026.20260911-1) over (2026.20260527-1)…

Progress: [ 51%] [                                                    ]
```

```
Session  Actions  Edit  View  Help
Processing triggers for sgml-base (1.31+nmu1)…
Processing triggers for debianutils (5.24)…
Processing triggers for wordlists (2026.2.0)…
Processing triggers for fontconfig (2.17.1-5)…
Processing triggers for desktop-file-utils (0.28-1)…
Processing triggers for initramfs-tools (0.151)…
update-initramfs: Generating /boot/initrd.img-7.1.5+kali-amd64
Processing triggers for hicolor-icon-theme (0.18-2)…
Processing triggers for doc-base (0.11.2)…
Processing 43 changed doc-base files, 3 added doc-base files...
Setting up docbook-xml (4.5-14)…
Processing triggers for libc-bin (2.43-6)…
Processing triggers for ca-certificates-java (20260311)…
done.
Setting up openjdk-25-jre:amd64 (25.0.4.1+1-1)…
Processing triggers for ca-certificates (20260816)…
Updating certificates in /etc/ssl/certs...
0 added, 0 removed; done.
Running hooks in /etc/ca-certificates/update.d...
done.
Processing triggers for php8.4-cli (8.4.24-1)…
Processing triggers for libapache2-mod-php8.4 (8.4.24-1)…
Processing triggers for sgml-base (1.31+nmu1)…
Processing triggers for ca-certificates-java (20260311)…
done.

┌──(kali㉿kali)-[~]
└─$
```

```
┌──(kali㉿kali)-[~]
└─$ apt list --upgradable
network-manager-applet/kali-rolling 1.36.0-5 amd64 [upgradable from: 1.36.0->
nm-connection-editor/kali-rolling 1.36.0-5 amd64 [upgradable from: 1.36.0-4]
openssh-client-gssapi/kali-rolling 1:10.5p1-1 all [upgradable from: 1:10.3p1>
python3-setuptools/kali-rolling 84.0.0-2 all [upgradable from: 78.1.1-0.1]
lines 1-4/4 (END)
```

Checking OS and Kernal version

```
┌──(kali㉿kali)-[~]
└─$ uname -r
7.1.5+kali-amd64

┌──(kali㉿kali)-[~]
└─$ lsb_release -a
No LSB modules are available.
Distributor ID: Kali
Description:    Kali GNU/Linux Rolling
Release:        2026.3
Codename:       kali-rolling

┌──(kali㉿kali)-[~]
└─$
```